Education and Employment

- J.J. Sylvester Assistant Professor at **Johns Hopkins University**, July 2024 – July 2027
- PhD in mathematics at **University College London**, October 2020 – July 2024
- Master of mathematics (MMath) at **The University of Warwick**, October 2016 – July 2020

Research Interests

- Calculus of Variations, Elliptic and Parabolic Partial Differential Equations, Geometric Measure Theory

Research

- *Isoperimetry of singular CMC hypersurfaces*
  With Gongping Niu
  Submitted for publication
- *Isoperimetry by stretching*
  With Gongping Niu
  International Mathematics Research Notices, volume 2026, issue 18
- *Generic regularity of isoperimetric regions in dimension eight*
  With Gongping Niu and Davide Parise
  Submitted for publication
- *Gradient flow of phase transitions with fixed contact angle*
  With Mayu Takada, Yoshihiro Tonegawa, and Myles Workman
  Interfaces and Free Boundaries, published online first
- *On isolated singularities and generic regularity of min-max CMC hypersurfaces*
  With Costante Bellettini
  The Journal of Geometric Analysis, volume 35, article number 126

Teaching

- Course leader and instructor at **Johns Hopkins University**:
  - Independent reading courses, Spring and Summer 2026
  - Partial Differential Equations II, Spring 2026
  - Partial Differential Equations I, Fall 2025
  - Honors Analysis I, Fall 2025
  - Calculus on Manifolds, Spring 2025 and Spring 2027
  - Calculus I (Physical Sciences and Engineering), Fall 2024 and Fall 2026
- Teaching assistant at **University College London**, 2020 – 2024:
  - Foundations, Mathematics for Science, Measure Theory, Probability Theory
- Undergraduate supervisor at **The University of Warwick**, 2019 – 2020:
  - Abstract Algebra, Analysis I and II, Differential Equations, Foundations, Geometry & Motion

Research with Students

- *Formalising linear elliptic PDE theory in Lean*
  With Alejandro José Soto Franco
  Submitted for publication

Invited Talks and Research Visits

- *University of Maryland Informal Geometric Analysis Seminar*, September 2026
- *University of Cambridge Geometric Analysis Seminar*, July 2026
- *Caltech Geometry & Topology Seminar*, May 2026
- *Stanford Geometry Seminar*, May 2026
- *University of Chicago Geometric Analysis Seminar*, May 2026
- *AMS Spring Southeastern Sectional Meeting*, March 2026
- *Notre Dame Geometry and Analysis Seminar*, March 2026
- *University of Tennessee, Knoxville Geometric Analysis Seminar*, October 2025
- *Brown University Topics in Differential Geometry Workshop*, June 2025
- *NCTS Differential Geometry Seminar*, May 2025
- Invited by Ulrich Menne to the *National Taiwan Normal University* for two weeks, May 2025
- *University of Tokyo Geometric Analysis Seminar*, May 2025
- *Keio University Nonlinear Analysis Seminar*, May 2025
- *Institute of Science Tokyo Analysis Seminar*, May 2025
- Invited by Yoshihiro Tonegawa to the *Institute of Science Tokyo* for one week, May 2025
- *Johns Hopkins Analysis & PDE Seminar*, September 2024
- Invited by Martin Li to the *Chinese University of Hong Kong* for one week, March 2024
- *University of Tokyo Analysis Seminar*, March 2024
- Invited by Yoshihiro Tonegawa to the *Tokyo Institute of Technology* for two weeks, March 2024
- *King's College London Analysis Seminar*, February 2024
- *Chinese University of Hong Kong Joint Geometric Analysis Seminar*, December 2023

Awards

- Joel Dean Excellence in Teaching Award at Johns Hopkins University, 2025 – 2026
- Joel Dean Excellence in Teaching Award at Johns Hopkins University, 2024 – 2025
- PhD studentship funded by the Engineering and Physical Sciences Research Council, 2020 – 2024
- For *Early Career Researchers in Mathematics 2022*:
  - Heilbronn Institute for Mathematical Research, £7000
  - London School of Geometry and Number Theory, £6200
  - London Mathematical Society, Scheme 8 - Postgraduate Research Conference Grants, £2500

Organisation and Service

- Graduate board oral examination committee chair at Johns Hopkins University, September 2026
- Organiser of the *Johns Hopkins Geometric Analysis Seminar*, since Spring 2025
- Thesis defence committee at Johns Hopkins University, June 2025
- Department representative for early career researchers at University College London, 2022 – 2024
- Co-organiser for *Early Career Researchers in Mathematics*, June 2022
- Co-organiser for the *Warwick Imperial Autumn Meeting*, November 2019
- Peer review for *Journal of Differential Equations* and *Journal of Differential Geometry*

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